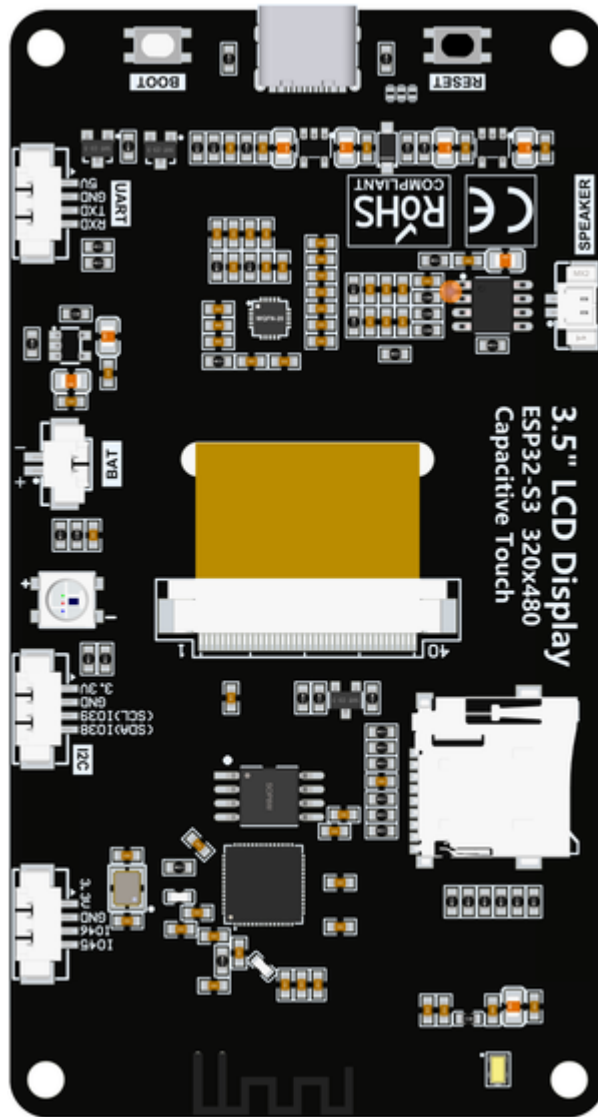


3.5inch ESP32-S3 Display - LCD wiki

lcdwiki.com/3.5inch_ESP32-S3_Display

Product image





Product introduction

- Using ESP32-S3 chip, convenient development, and sufficient development resources.
- 3.5-inch color screen with a resolution of 320*480, supporting up to RGB565, offering a rich display colors.
- Rich interfaces, convenient to connect various peripherals (IIC, UART, etc.)
- Support external speakers, built-in microphone, can play and receive audio
- Equipped with RGB three-color indicator lights, the status indication is rich.
- Integrated display and touch, with built-in capacitive touch screen, convenient for human-machine interaction.
- Standard TYPE-C interface, convenient for program download and power supply
- Built-in micro TF card slot, convenient for storage expansion
- Support external lithium battery, lightweight and portable
- Built-in battery charging management circuit ensures safe battery charging and discharging.
- Provide a wealth of example programs for easy learning.
- Provide underlying technical support for driving, and update WIKI materials online.

- The module aging test can reach military-grade standards with multiple detections, supporting long-term stable operation.
- Support for Xiaozhi AI voice chat;

Product parameters

ESP32 main control parameters

Name	parameter
Module	ESP32-S3
CPU	Xtensa LX7 32-bit dual-core processor
Clock frequency	240MHz (MAX)
Storage	384 KB ROM + 512KB SRAM +16 KB RTC SRAM + 8M OPI PSRAM + 16M QSPI Flash (N16R8)
WIFI	2.4GHz、 802.11b/g/n
Bluetooth	Bluetooth V5.0 BR/EDR and Bluetooth LE standard
Working voltage	3.0~3.6(V)

Display touch integrated LCD screen parameters

Name	Parameters
Screen size	3.5 inch
Screen type	IPS TFT
Screen resolution	320xRGBx480(pixels)
Effective display area	48.96(W)x73.44(H)
Number of colors	65K (RGB565
Driver IC	ST77922

Display interface	QSPI(Received on ESP32-S3)
Pixel Size	0.153(H)x0.153(mm)
Viewing angle	ALL 0'CLOCK
Back brightness (typical value)	300 cd/m ²
Backlight lamp type	White LED*6
Effective touch area	54.50(W)x83.00(H)(mm)
Touch screen type	Capacitive touch screen
resolution	320x480(pixels)
Visible window size	49.96±0.15(W)x74.44±0.15(H)(mm)
Communication interface	IIC
Operating temperature	-30~80(°C)
Storage temperature	-30~80(°C)

Size Parameters

Name		Parameters
LCD screen size	54.50±0.2(W)x83.00±0.2(H)x3.2(D)	(mm)
TOUCH screen size	54.50±0.2(W)x83.00±0.2(H)x1.0(D)	(mm)
Module size	54.50(W)x101.50(H)x10.50(D)	(mm)

Battery charging parameters

Name		Parameters
Charging voltage	Range: 4.2~6.5 Typical value: 5.0	

Charging current	Maximum value: 500mA Module actual: 290mA
Charging saturation voltage	4.24V
Charging temperature	Module actual maximum value: 62°C
Charging battery specifications	3.7V lithium polymer battery

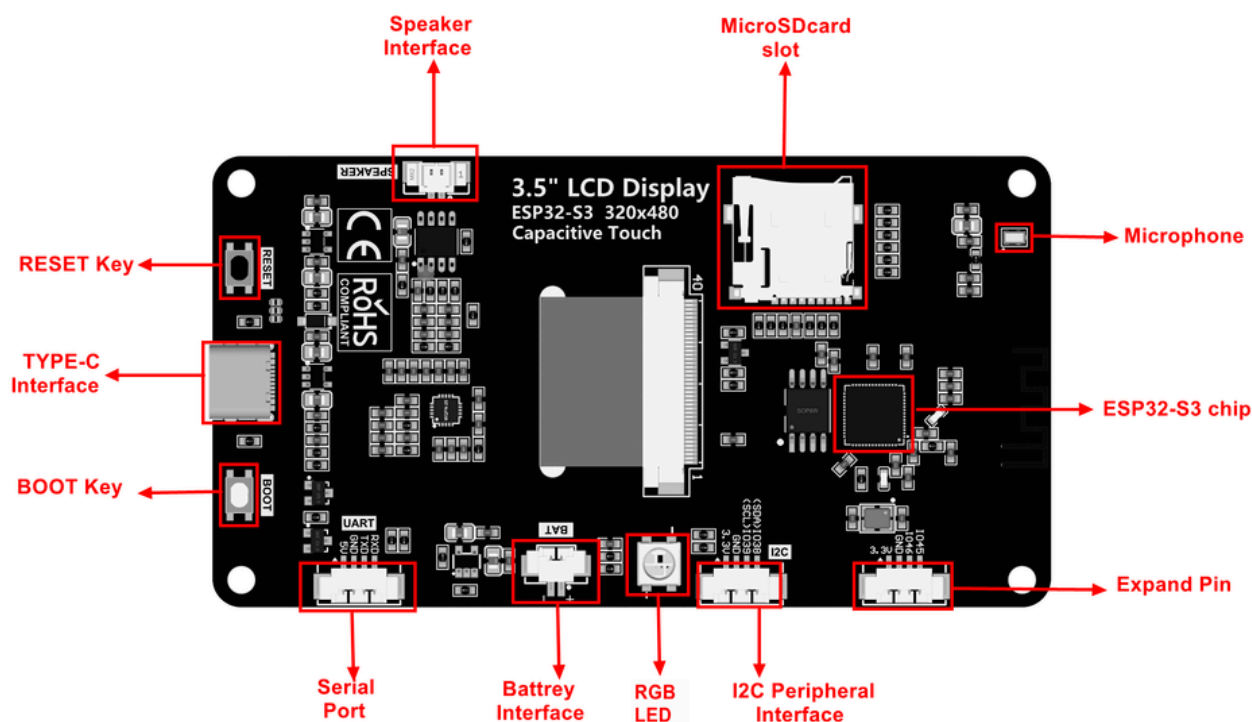
Electrical Parameters

Name	Parameters
Working voltage	5.0V
Backlight current	120mA
Backlight brightness (actual value)	300cd/m ²
Total current	ESP32-S3 reset: 0 Only display works:198 Display, speaker, battery charging all work:401
Power	0.96 (Only the display works) 1.76 (Display, speaker, battery charging all work)

Basic Parameters

Name	Parameters
SKU	The product comes with a speaker : ES3C35P Product does not come with a speaker : ES3C35P-NS
Power interface	USB(Type-C)
Weight (including packaging)	

PRODUCT INTERFACE



Interface Function Description

Interface name	Function Description
ESP32-S3	Main control of the display module, working together with the peripheral circuit, is used to control on-board peripherals and external peripherals.
MicroSD card slot	Insert a SD card to expand storage space, such as storing large data content such as font libraries, images, audio files, etc.
RGB tricolor light	Contains LEDs of three colors: red, green, blue. It has a built-in control IC and only needs one IO port to control. It can be used for LED testing and status indication.

Serial port	1.25mm4P socket. It can be used for serial port debugging, downloading, and communication. An external USB to serial port module is required.
Battery interface	1.25mm 2P socket. Used to connect to a 3.7V polymer lithium battery, charge the battery through the battery management circuit, and can also be used for battery power supply. Note the positive and negative terminals of the interface.
BOOT button	Used to enter the download mode or key test. Press and this key to power up, then release to enter the download mode, or after powering up, press and hold this key, then press the RESET key, release the RES key and then release this key, you can also enter the download mode. When you don't need to enter the download mode, this key can be used as a normal key
TYPE-C interface	It is used for module power supply and program download, debugging. This interface is connected to the internal USB bus of ESP32-S3, which can generate a USB serial port. Note: win10 and above systems support USB virtual serial port function. If you use this interface on other versions of the system, you need to solve the problem USB virtual serial port not working by yourself.
RESET button	For ESP32-S3 main control and LCD reset, press to reset low level.
Expansion pin	1.25mm 4P socket. Two pins are GPIO45, GPIO46. It can be used to connect other peripherals.
Loudspeaker interface	1.25mm P socket. Used to connect the speaker to play audio. Used to connect the speaker to play audio (maximum support 1.5W (8Ω) or 2W4Ω) speaker).
I2C peripheral interface	Downward-facing MEMS silicon microphone. Used to capture external sound. Can be used for those projects that require audio input function.

ESP32-S3 pin allocation

ESP32-S3		
Device	pin	pin allocation

LCD	IO10	LCD screen chip select control signal, low level active
	IO12	LCD QSPI bus clock signal
	IO11	LCD QSPI data bus D0
	IO13	LCD QSPI data bus D1
	IO14	LCD QSPI data bus D2
	IO9	LCD QSPI data bus D3
	EN	LCD screen reset control signal, low level reset (share reset pin with ESP32-S3 main control)
	IO41	LCD screen backlight control signal (level turn on back-light, low level turn off backlight)
Capacitive touch screen	IO38	Capacitive touch screen I2C bus data signal (share with audio codec IC and extended I2C)
	IO39	Capacitive touch screen I2C bus clock signal (share with audio codec IC and extended I2C)
	IO48	Capacitive touch screen reset control signal, low level
	IO47	Capacitive touch screen interrupt input signal, input low level when touch event occurs.
RGB light	IO40	RGB three-color LED light control signal with built-in control IC. Different timing can be input to control the internal red, green and blue three kinds of beads respectively.
MicroSD	IO5	SD card SDIO bus clock signal
	IO4	SD card SDIO bus command signal
	IO6/7/2/3	SD SDIO bus data signal DATA0~DATA3

Audio	IO1	Audio power amplifier IC enable pin, low level enable
	IO17	Audio I2S bus master clock signal
	IO18	Audio I2S bus bit clock signal
	IO15	Audio I2S bus bit output data signal
	IO21	Audio I2S bus left and right channel selection signal. High level: right channel; low level: left channel
	IO16	I2S bus bit input data signal
	IO38	Audio codec IC's I2C bus data signal (share with capacitive touch screen and extended I2C)
	IO39	Audio codec's I2C bus clock signal (share with capacitive touch screen and extended I2C)
Button	IO0	Download mode selection button (hold this button to power up, then release enter download mode)
	EN	ESP32-S3 reset button, low level reset (share with LCD screen reset)
Serial por	RXD0(IO43)	ESP32-S3 serial port 0 receive signal interface (if the serial port communication function is not used, it can be used as a IO port)
	TXD0(IO44)	ESP32-S3 serial port 0 receive signal interface (if the serial port communication function is not used, it can be used as a general port)
Battery	IO8	Battery voltage ADC value acquisition input signal
Expansion pin	IO45	ports that can be used to connect peripherals
	IO46	ports that can be used to connect peripherals
	GND	GND
	VCC	3.3V

I2C peripheral interface	IO38	Expansion interface I2C bus data signal (shared with capacitive touch screen and audio codec IC, can be used a general IO when touch and audio functions are not used, otherwise it can only be used as an I2C interface)
	IO39	Expansion interface I2C bus clock signal (with capacitive touch screen and audio codec IC, can be used as a general IO when touch and audio functions are not used, otherwise it can only be used as an I2C interface)

Quick usage guide

[3.5inch_ESP32-S3_Quick_Start](#)

[3.5inch_ESP32-S3_Quick_Start_Manual](#)

Data pack download

[3.5-inch ESP32-S3-ST77922 Driver Display Module Data Package \(Extract Code: an79\)](#)

[3.5-inch ESP32-S3-ST77922 Driver Display Module Data Package](#)

Product documentation

Specifications

Product Specifications

[3.5inch_ESP32-S3_Specification](#)

User manual

[3.5inch_IPS_ESP32-S3_User_Manual](#)

Size Drawing

[ES3C35P_Size](#)

Schematic

[3.5inch_ESP32-S3_Display_Hardware schematic diagram](#)

IO Resource Allocation Table

[ESP32-S3-I/O resource allocation table](#)

Reference Materials

Development environment setup

[ESP32 Arduino IDE Development environment setup](#)

[ESP32 MicroPython Development environment setup](#)

[ESP32 ESP-IDF Development environment setup](#)

[ESP-IDF LVGL porting construction for ESP32](#)

Tool Software

[Flash_Download_Tool \(Official website download\)](#) [Flash_Download_Tool](#)

[JPGCompact](#)

[TCP_UDP test tool](#)

[USB-SERIAL_CH340](#)

[Serial debugging assistant](#)

[Network debugging assistant](#)

[esptouch](#)

[PCtoLCD2002](#)

[Image2Lcd](#)

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